Multiobjective Optimization

A-priori Methods: Utility Functions and Weighting Approaches

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A-priori Methods - Recap

Key Concepts:

- Preferences articulated **before** solution
- Goal: Identify **best-compromise** directly
- Two main approaches covered:
  1. Multiattribute Utility Functions
  2. Prior Assessment of Weights

Today:

- Deeper dive into utility functions
- Advanced weighting approaches
- Geometric methods (Goal Programming)

Theoretical Foundation

General Form:

$$U[Z_1(\mathbf{x}), Z_2(\mathbf{x}), \ldots, Z_p(\mathbf{x})]$$

Key Axioms (von Neumann & Morgenstern):

1. **Completeness:** All alternatives can be ordered
2. **Transitivity:** If $A \succ B$ and $B \succ C$, then $A \succ C$
3. **Continuity:** Small changes in objectives give small changes in utility
4. **Independence:** Preferences over lotteries are consistent

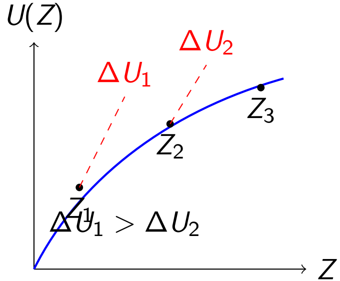
Behavioral Assumptions

For Maximization:

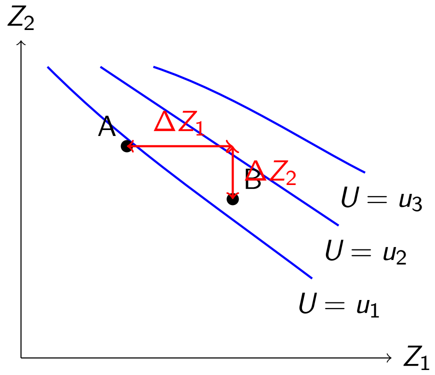
$$\frac{\partial U}{\partial Z_k} > 0 \quad \forall k \quad \text{(Monotonicity)}$$

$$\frac{\partial^2 U}{\partial Z_k^2} < 0 \quad \forall k \quad \text{(Diminishing Marginal Utility)}$$

Geometric Interpretation:



Isopreference Curves



Note: MRS decreases as Z_1 increases (diminishing marginal utility)

Decomposition Approach

General Form:

$$U(Z_1, Z_2, \ldots, Z_p) = \sum_{k=1}^{p} U_k(Z_k)$$

Example (Briskin, 1966):

$$Z_2 = \rho(1 - e^{\sigma Z_1})$$

Where:

- ρ, σ are constants to be estimated
- MRS: $\mathrm{MRS}_{12} = \rho\sigma e^{\sigma Z_1}$
- Estimate from decision maker's tradeoff descriptions

Advantage: Easier to assess than full utility function

More General Form

Multiplicative Form (Keeney, 1974):

$$1 + U(Z_1, Z_2) = [1 + \alpha U_1(Z_1)][1 + \beta U_2(Z_2)]$$

Special Cases:

- If $\alpha = \beta = 0$: **Additive**
- If $\alpha \neq 0$ or $\beta \neq 0$: **Multiplicative**

Interpretation:

- Captures preference interactions
- More realistic but harder to estimate
- Requires additional independence assumptions

Practical Challenges

Methods for Assessment:

1. **Lottery Method:**
   - Decision makers choose between lotteries
   - Infer utility values from choices
2. **Direct Rating:**
   - Rate alternatives on 0-100 scale
   - Fit curve to points
3. **Tradeoff Method:**
   - Ask about acceptable tradeoffs
   - Estimate MRS at different points

Key Issue: Decision makers may not have well-defined preferences!

Direct Best-Compromise Search

Problem:

$$\text{maximize } U[Z_1(\mathbf{x}), Z_2(\mathbf{x})]$$

$$\text{s.t. } \mathbf{x} \in \mathbf{F}_d$$

Algorithm (for 2 objectives):

1. Solve weighted problem with $\alpha = 0$: get $\mathbf{x}^0$
2. Increase α parametrically: get $\mathbf{x}^1, \mathbf{x}^2, \ldots$
3. Compare $U(\mathbf{x}^i)$ and $U(\mathbf{x}^{i+1})$
4. When utility decreases, best-compromise lies between $\mathbf{x}^i$ and $\mathbf{x}^{i+1}$
5. Search along line segment for maximum

Step-by-Step

Step 1: $i = 0, j = 1, \alpha = 0$
Solve: maximize $Z_2(\mathbf{x})$ with $\mathbf{x} \in \mathbf{F}_d$

Step 2: Increase α until new solution $\mathbf{x}^{i+1}$

Step 3: Compare utility:

$$U[Z(\mathbf{x}^i)] \quad \text{vs} \quad U[Z(\mathbf{x}^{i+1})]$$

Step 4: If utility increased: $j = i, i = i + 1$, go to Step 2

Step 5: If utility decreased: best-compromise lies between $\mathbf{x}^i$ and $\mathbf{x}^j$

Key Idea:

"Find the feasible solution closest to the decision maker's goals."

Goals: Target values G_k for each objective

Formulation:

$$\text{minimize } d_1 = \sum_{k=1}^{p} |G_k - Z_k(\mathbf{x})|$$

Interpretation:

- Minimizes total deviation from goals
- Uses d_1 metric (Manhattan distance)
- Can be formulated as linear program

Converting to Linear Program

Minimize:

$$\sum_{k=1}^{p}(d_k^+ + d_k^-)$$

Subject to:

$$Z_k(\mathbf{x}) - d_k^+ + d_k^- = G_k \quad \forall k$$

$$d_k^+, d_k^- \geq 0 \quad \forall k$$

$$\mathbf{x} \in \mathbf{F}_d$$

Where:

- d_k^+: Positive deviation (above goal)
- d_k^-: Negative deviation (below goal)

Adding Priorities

Weighted Form:

$$\text{minimize} \sum_{k=1}^{p} (w_k^+ d_k^+ + w_k^- d_k^-)$$

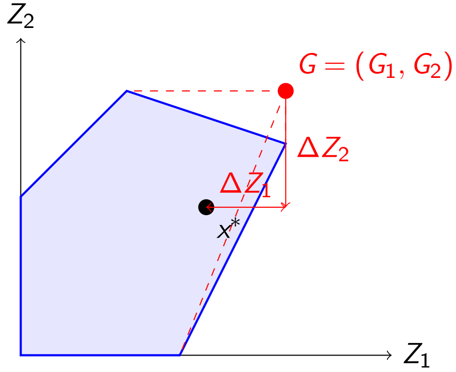
Special Cases:

1. **One-sided:** $w_k^+ = 0$ (only below goal matters)
2. **Lexicographic:** $w_k \gg w_j$ (some goals absolutely more important)
3. **Symmetric:** $w_k^+ = w_k^-$ (both directions equally important)

Example: Water quality - Dissolved oxygen ≥ 5 mg/L

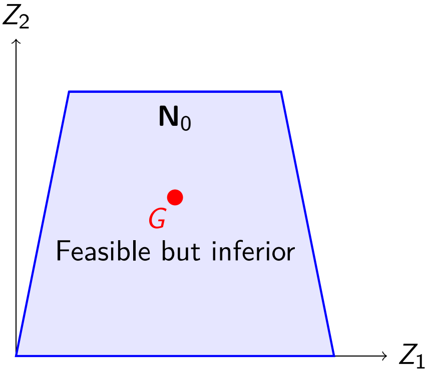
- Only negative deviations matter ($w_k^+ = 0$)

Finding the Closest Point



Goal: Minimize Manhattan distance to G

Warning: May Find Inferior Solutions



Problem: If goals are feasible, solution may be inferior!
Solution: Check noninferiority of goal programming solution

Alternative Distance Metrics

Ideal Solution:

$$Z_k^* = \max_{\mathbf{x} \in \mathbf{F}_d} Z_k(\mathbf{x})$$

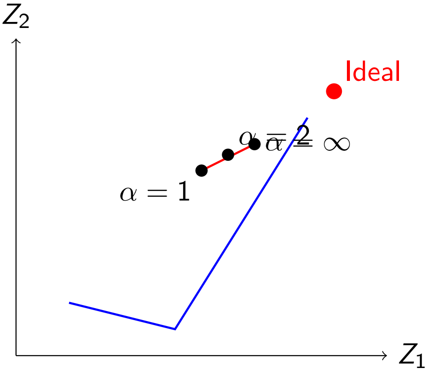
General Distance Metric:

$$d_\alpha = \left\{ \sum_{k=1}^{p} (Z_k^* - Z_k(\mathbf{x}))^\alpha \right\}^{1/\alpha}$$

Special Cases:

- $\alpha = 1$: Sum of deviations (equal weighting)
- $\alpha = 2$: Euclidean distance
- $\alpha = \infty$: Minimize maximum deviation (minimax)

Range of Solutions



Compromise Set: Solutions for $1 \leq \alpha \leq \infty$

Hybrid Approach

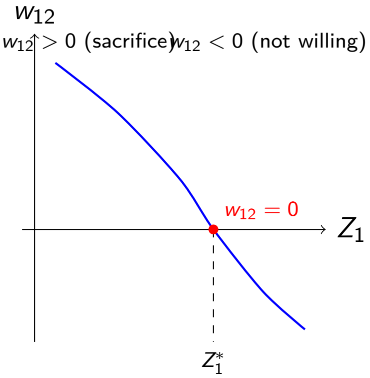
Key Idea:

"Ask decision makers to evaluate tradeoffs, then find where MRS = MRT."

Procedure:

1. Generate noninferior solutions (constraint method)
2. Compute tradeoffs $t_{12}(Z_1)$
3. Ask decision maker to assign "surrogate worth" to each tradeoff
4. Find where surrogate worth = 0 (point of indifference)

Output: Best-compromise solution



Interpretation:

- $w_{12} > 0$: DM willing to sacrifice Z_2 for Z_1
- $w_{12} < 0$: Not willing to sacrifice
- $w_{12} = 0$: Indifferent (best-compromise)

Numerical Example

Sample problem (Cohon, 1978):

Z_1	Z_2	$\lvert t_{12} \rvert$	w_{12}
7.3	11.5	0.34	$+3.65$
22.0	4.5	0.79	-0.75

Result:

$$Z_1^* = 19.6, \quad Z_2^* = 6.5$$

Compare: Actual best-compromise is $(14.4, 10.3)$
Note: Approximation error due to linear approximation of tradeoff function

Summary and Tradeoffs

Method	**Information Required**	**Computational Cost**
Prior Weights	Weights	Low
Utility Function	MRS, lottery choices	Medium
Goal Programming	Goals, priorities	Low
SWT	Tradeoff evaluations	Medium
Geoffrion	MRS, step size choices	Medium-High

Key Tradeoff:

"Simplicity of information vs. accuracy of preference representation"

When to Use Which Method

Consider:

1. **Decision Maker Availability:**
   - Limited: Use generating methods or SWT
   - Extensive: Use utility functions or iterative methods
2. **Preference Articulation:**
   - Easy: Use weights or goals
   - Complex: Use utility functions
3. **Computational Budget:**
   - Limited: Use prior weights or goal programming
   - Generous: Use Geoffrion or SWT

Key Takeaways

A-priori Methods - Advanced:

- **Utility Functions:** Complete but difficult to estimate
  - Additive, multiplicative, and other decompositions
- **Geoffrion's Algorithm:** Systematic search using MRS
- **Goal Programming:** Distance from goals (may find inferior solutions!)
- **Compromise Programming:** Distance from ideal (compromise set)
- **SWT:** Hybrid method using tradeoff evaluations

Next: Multiple-Decision-Maker Methods

Multiple-Decision-Maker Methods

Topics:

1. Aggregation of Individual Preferences
   - Welfare economics overview
   - Social welfare functions
2. Counseling Individual Decision Makers
3. Predicting Political Outcomes
   - Game theory
   - Vote-trading models

Key Question: How do we handle multiple conflicting decision makers?

References

COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.